Module 02

Why DBMS?

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Module Objectives

- To understand the need for a DBMS from a historical perspective

Module Outline

- Evolution of data management practices

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- History of DBMS

Evolution of Data Management

Data Management

Definition

Management of Data or Records is a basic need for human society:

- Storage

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There have been two major approaches in this practice:

- Physical

Data Management: Physical

Physical Data or Records management, more formally known as **Book Keeping**, has been using physical ledgers and journals for centuries.

The most significant development happened when Henry Brown, an American inventor, patented a 'receptacle for storing and preserving papers' on November 2, 1886.

Herman Hollerith adapted the punch cards used for weaving looms to act as the memory for a mechanical tabulating machine, in 1890.

Data Management: Electronic

Growth with Technology

Electronic Data or Records management moves with the advances in technology especially of memory, storage, computing, and networking.

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- **2010s:** Data Science started riding high

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Not only small shops but large organizations also used to maintain their transaction details in book registers.

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Spreadsheet Softwares like Google Sheets: Due to the disadvantages of maintaining ledger registers, organizations dealing with huge amount of data shifted to using spreadsheet softwares for maintaining their records in files.

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Mostly useful for single user or small enterprise applications.

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- **Consistency:** Not guaranteed but spreadsheets are less prone to mistakes than registers.

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Database Management Systems

The above limitations of filesystems paved the way for a comprehensive platform dedicated to management of data – the **Database Management Systems**.

History of DBMS

History of Database Systems

1950s - 60s

- Magnetic tapes
- Sequential access
- Punched cards

Late 60s - 70s

- Hard disks (direct access)
- Network & Hierarchical models
- Ted Codd (Relational model)

1980s

- SQL industrial standard
- Parallel & distributed DBs
- Object-oriented DBs

1990s

- Data warehousing
- Data mining
- Emergence of Web commerce

History of Database Systems (2)

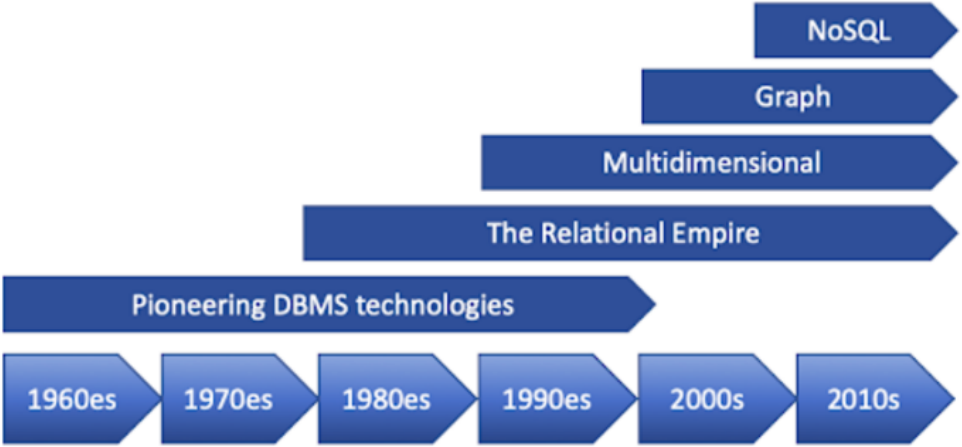
Early 2000s

- XML and XQuery standards
- Automated database administration

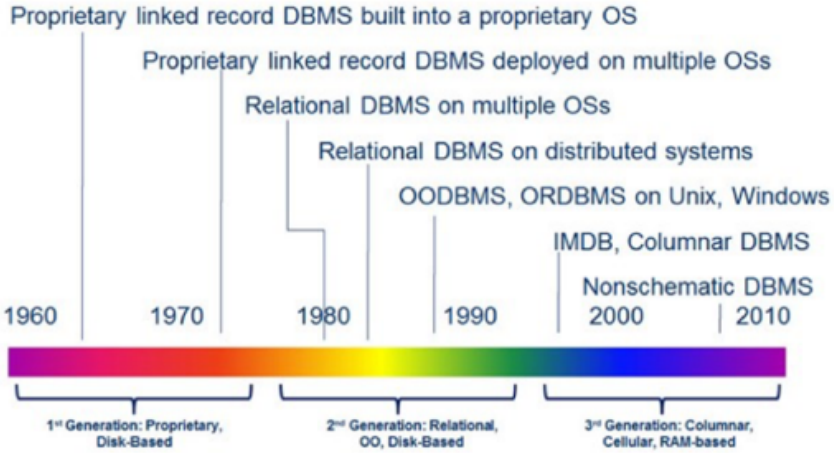
Later 2000s

- Giant data storage systems
  - Google BigTable
  - Yahoo PNuts
  - Amazon Dynamo

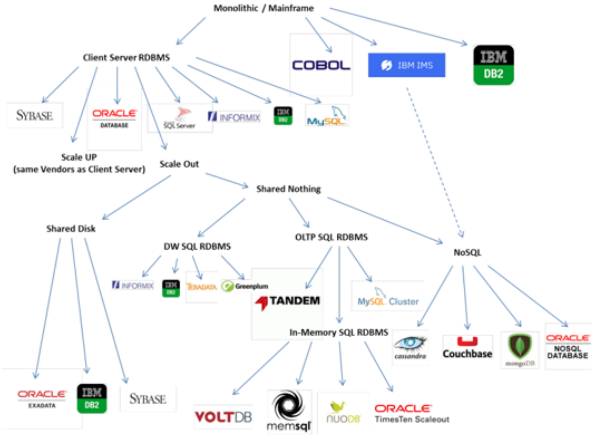
History (3): Evolution of Data Models



History (4): Evolution of DB Technology



History (5): Evolution of DB Architecture



Module Summary

- Walk through of evolution of Data and Records Management
- History of DBMS

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